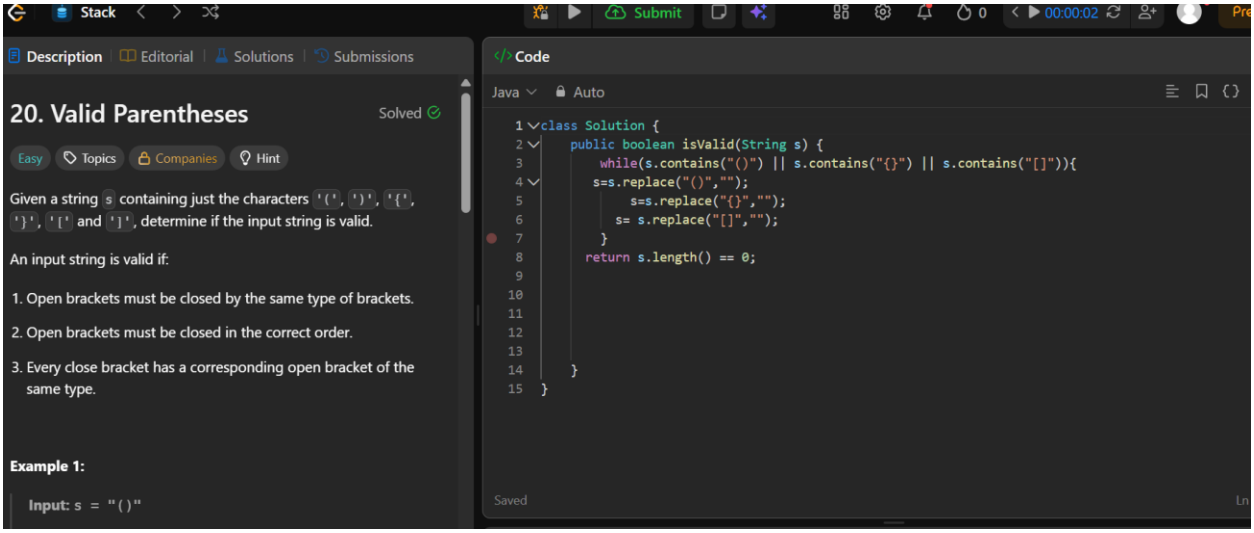
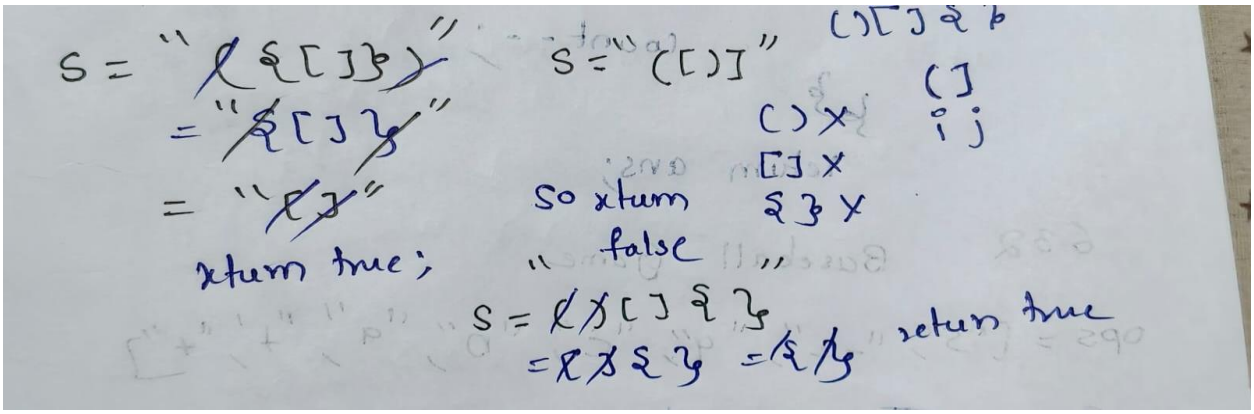
T. DHANA LAXMI
2505A41L46

1.valid parentheses LC-20

https://leetcode.com/problems/valid-parentheses/





2.min stack LC-155

https://leetcode.com/problems/min-stack

Stack

Submit

00:00:02

Description | Editorial | Solutions | Submissions

155. Min Stack

Solved

Medium | Topics | Companies | Hint

Design a stack that supports push, pop, top, and retrieving the minimum element in constant time.

Implement the `MinStack` class:

- `MinStack()` initializes the stack object.
- `void push(int value)` pushes the element `value` onto the stack.
- `void pop()` removes the element on the top of the stack.
- `int top()` gets the top element of the stack.
- `int getMin()` retrieves the minimum element in the stack.

You must implement a solution with `O(1)` time complexity for each function.

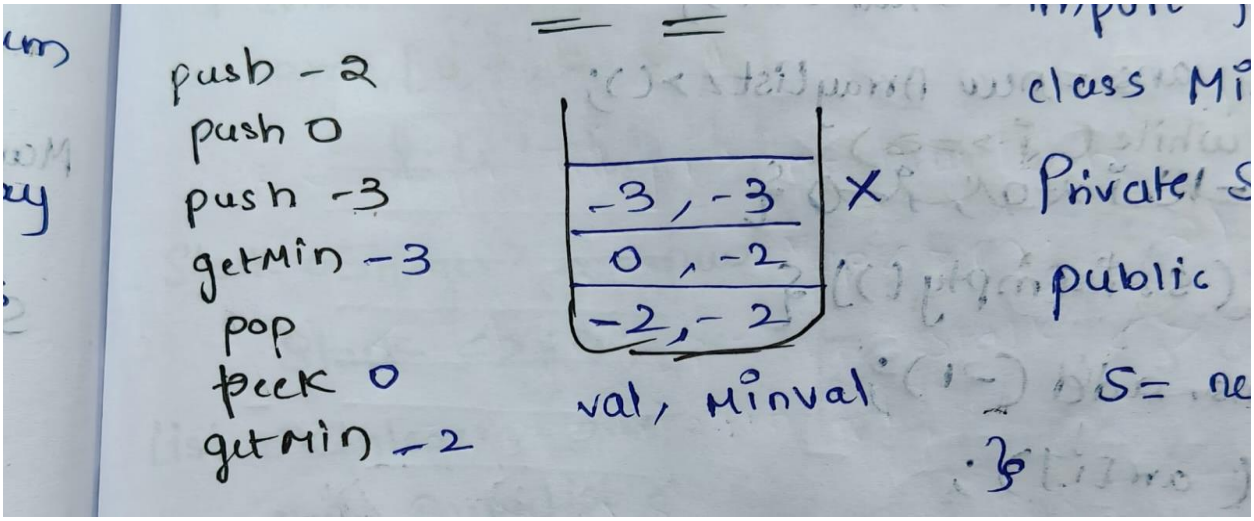
Code

Java | Auto

```java
class MinStack {
    private Stack<int[]> s; // {val, minVal}
        public MinStack() {
        s = new Stack<>();
    }public void push(int val) {
        if (s.isEmpty()) {
            s.push(new int[]{val, val});
        } else {
            int minVal = Math.min(val, s.peek()[1]);
            s.push(new int[]{val, minVal});
        }
    }public void pop() {
        s.pop();
    }public int top() {
        return s.peek()[0];
    }public int getMin() {
        return s.peek()[1];
    }
}
```

Saved

Testcase | Test Result



3.next greatest element LC-496

https://leetcode.com/problems/next-greater-element-i/

Stack

Submit

00:00:02

Description | Editorial | Solutions | Submissions

496. Next Greater Element I

Solved

Easy | Topics | Companies

The **next greater element** of some element `x` in an array is the **first greater** element that is **to the right** of `x` in the same array.

You are given two **distinct 0-indexed** integer arrays `nums1` and `nums2`, where `nums1` is a subset of `nums2`.

For each `0 <= i < nums1.length`, find the index `j` such that `nums1[i] == nums2[j]` and determine the **next greater element** of `nums2[j]` in `nums2`. If there is no next greater element, then the answer for this query is `-1`.

Return *an array* `ans` *of length* `nums1.length` *such that* `ans[i]` *is the* ***next greater element*** *as described above.*

Example 1:

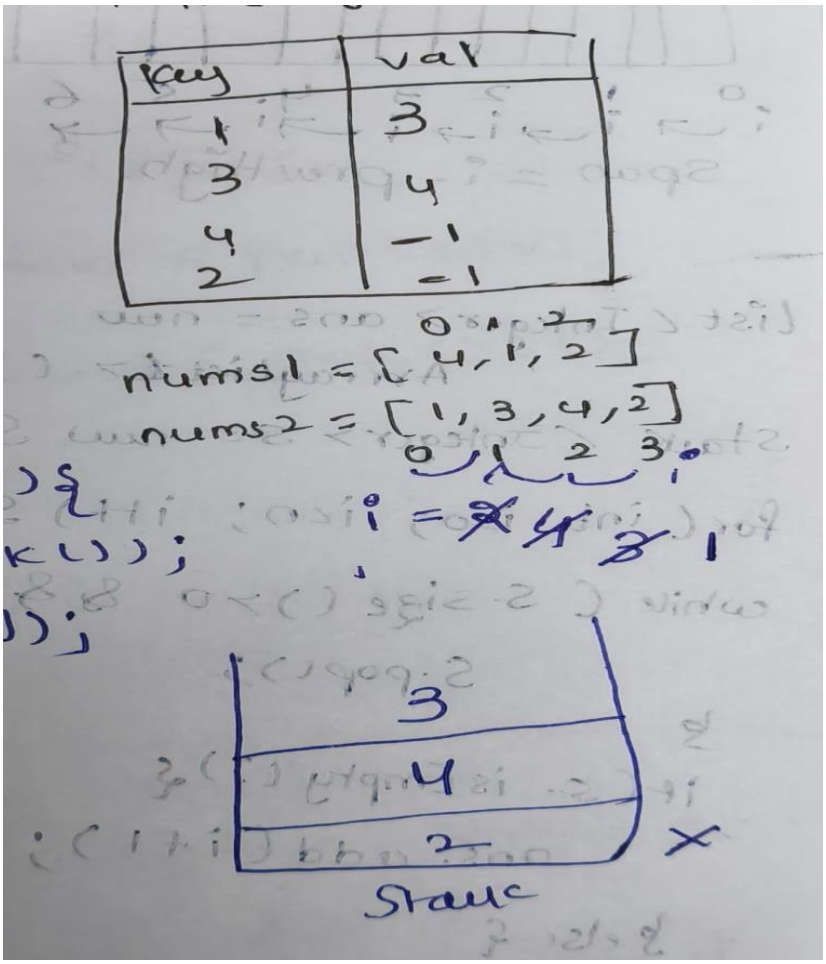
Code

Java | Auto

```java
class Solution {
    public int[] nextGreaterElement(int[] nums1, int[] nums2) {
      Map <Integer,Integer> m = new HashMap<>();
      Stack <Integer> s = new Stack<>();  int i = nums2.length-1;
      while(i >= 0){
        if(s.isEmpty()){
            m.put(nums2[i],-1);
            s.push(nums2[i]);
            i--;
        }else if(nums2[i] < s.peek()){
            m.put(nums2[i],s.peek());
            s.push(nums2[i]);
            i--;
        }else s.pop();
      } int ans[] = new int[nums1.length];
      for(int j=0; j<nums1.length; j++)  ans[j] = m.get(nums1[j]);
      return ans;
    }
}
```

Saving...

Testcase | Test Result



4. daily temperatures LC-739

https://leetcode.com/problems/daily-temperatures/

Stack

Submit

00:00:02

Description | Editorial | Solutions | Submissions

739. Daily Temperatures

Solved

Medium | Topics | Companies | Hint

Given an array of integers `temperatures` represents the daily temperatures, return *an array* `answer` *such that* `answer[i]` *is the number of days you have to wait after the* `i`th *day to get a warmer temperature.* If there is no future day for which this is possible, keep `answer[i] == 0` instead.

Example 1:

```
Input: temperatures = [73,74,75,71,69,72,76,73]
Output: [1,1,4,2,1,1,0,0]
```

Example 2:

```
Input: temperatures = [30,40,50,60]
Output: [1,1,1,0]
```

15.1K | 441 | 38 Online

Code

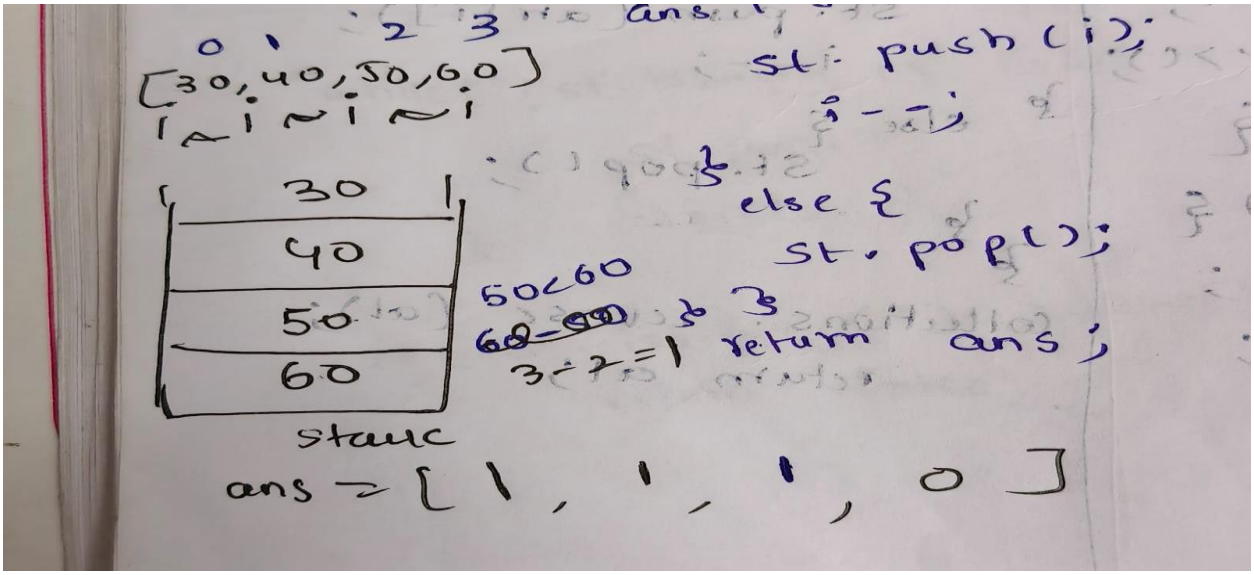
Java | Auto

```java
class Solution {
    public int[] dailyTemperatures(int[] arr) {
        Stack <Integer> st = new Stack <>();
        int n = arr.length;int i = n-1; int ans[] = new int[n];
      while(i >= 0){
        if(st.isEmpty()){
        ans[i] = 0;
        st.push(i);
        i--;
       }
        else if(arr[i] < arr[st.peek()]){
      ans[i] = st.peek() - i;
        st.push(i);
          i--;
        }
         else st.pop();
       }
     return ans;
    }
```

Saved

Testcase | Test Result

You must run your code first



5.evaluate reverse polish notation

https://leetcode.com/problems/evaluate-reverse-polish-notation/

150. Evaluate Reverse Polish Notation

Solved

Medium | Topics | Companies

You are given an array of strings `tokens` that represents an arithmetic expression in a Reverse Polish Notation.

Evaluate the expression. Return *an integer that represents the value of the expression.*

Note that:

- The valid operators are `'+'`, `'-'`, `'*'`, and `'/'`.
- Each operand may be an integer or another expression.
- The division between two integers always **truncates toward zero**.
- There will not be any division by zero.
- The input represents a valid arithmetic expression in a reverse

9K | 484 | 26 Online

Java | Auto

```java
class Solution {
    public int evalRPN(String[] tokens) {
        Stack <Integer> st = new Stack<>();
        String operators = "+-*/";
        for(String token:tokens){
            if(operators.indexOf(token) != -1){
            int b = st.pop();
             int a = st.pop();
             if(token.equals("+")) st.push(a+b);
             else  if(token.equals("-")) st.push(a-b);
              else if(token.equals("*")) st.push(a*b);
              else  st.push(a/b);
            }else{
                st.push(Integer.parseInt(token));
            }
        }
        return st.peek();
    }
}
```

Saved | Ln 4,

